

THE HONG KONG POLYTECHNIC UNIVERSITY
DEPARTMENT OF COMPUTING
EXAMINATION

Course : BSc Scheme in COMP-61031, Broad Discipline of COMP-61431,
BSc COMP & BBA Mgt-61032, BEng EIE-42470

Subject : COMP4431 Artificial Intelligence

Group : 1011

Session : 2016 / 2017 Semester I

Date : 17 December 2016

Time : 12:30 - 14:30

Time Allowed: 2 Hours

Subject Lecturer: Dr LIU Yan

This question paper has 5 pages.
(Some pages may be intentionally omitted.)

Instructions to Candidates:

This is a closed-book examination.

Answer ALL questions.

Calculator is permitted.

Do not turn this page until you are told to do so!

Section B

[80 Points]

Question 2. Answer the question in a short paragraph.

2.1. Could you give one example to explain the difference between forward chaining and backward chaining? [4]

2.2. As shown in Figure 1, five two-dimensional data points belonged to two categories (star and circle) are given for the training of support vector machine. Could you indicate support vectors and calculate the linear separator? [4]

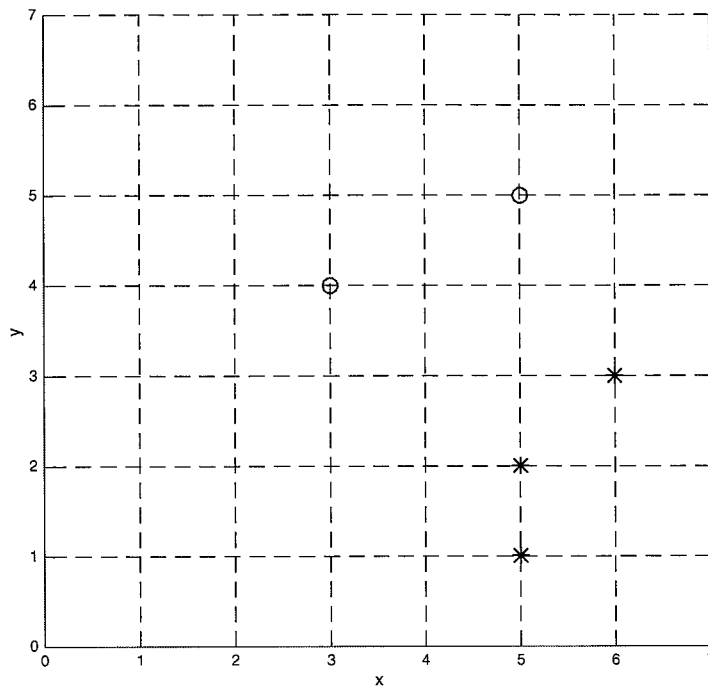


Figure 1. Figure of question 2.2

Question 3. As shown in Figure 2, we want to find a way to drive from City B to City G. We show the distance between two cities in the figure. We formulate the problem to the search tree. The heuristic information is provided as the straight line distance between two cities as shown below:

A-G: 30; B-G: 25; C-G: 15; D-G: 5; E-G: 10; F-G: 20; G-G: 0;

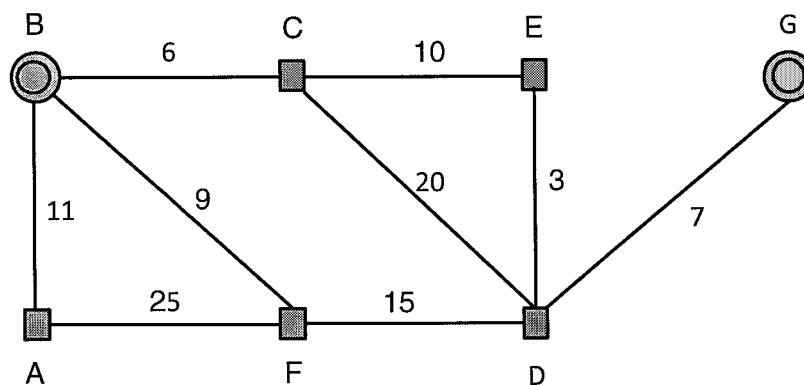


Figure 2. Figure of question 3

3.1. Solve the problem using breadth-first search. The target is to find a path from City B to City G. Draw the search tree and explain the order of the node checked or/and expanded. [4]

3.2. Solve the problem using depth-first search. The target is to find a path from City B to City G. Draw the search tree and explain the order of the node checked or/and expanded. [4]

- 3.3. Solve the problem using uniform-cost search. Draw the search tree and explain the order of the node checked or/and expanded. [4]
- 3.4. Solve the problem using greedy best-first search. Draw the search tree and explain the order of the node checked or/and expanded. [4]
- 3.5. Solve the problem using A* search. Draw the search tree and explain the order of the node checked or/and expanded. [4]
- 3.6. Does this question satisfy the requirement of admissible heuristic? Provide the detailed explanation of your conclusion by calculation. [4]

Question 4. We want to use neural networks in classification problem.

4.1. Eight 4-dimensional data points $(x_1, x_2, x_3, x_4)_i$ are given as follows: $\{(0.14, -0.54, 0.26, 0.33), (0.38, 0.26, 0.19, -0.09), (-0.42, -0.2, -0.47, -0.57), (0.34, -0.3, -0.3, -0.27), (-0.13, 0.05, -0.57, 0.24), (-0.04, 0.25, -0.25, -0.15), (-0.55, -0.42, 0.06, -0.27), (0.3, -0.48, 0.39, -0.06)\}$ The corresponding labels y_i are $\{-1, 1, -1, -1, 1, 1, -1, -1\}$. The learning rate η is set to 0.1. Please calculate the value of w_i according to the given perceptron learning algorithm. [10]

Perceptron Learning algorithm

1. Initialize a random w_0 as $(0.1, -0.02, -0.02, 0.07)$
2. For each (x_i, y_i) , η is the learning rate.
 - 2a: calculate the estimated label $y_i^{est} = \text{sgn}(w^T x_i)$
 - 2b: update the weight $w_{i+1} = w_i + \eta(y_i - y_i^{est})x_i$

4.2. Could you please write down the back propagation algorithm according to the given neural networks and activation function shown in Figure 3? [10]

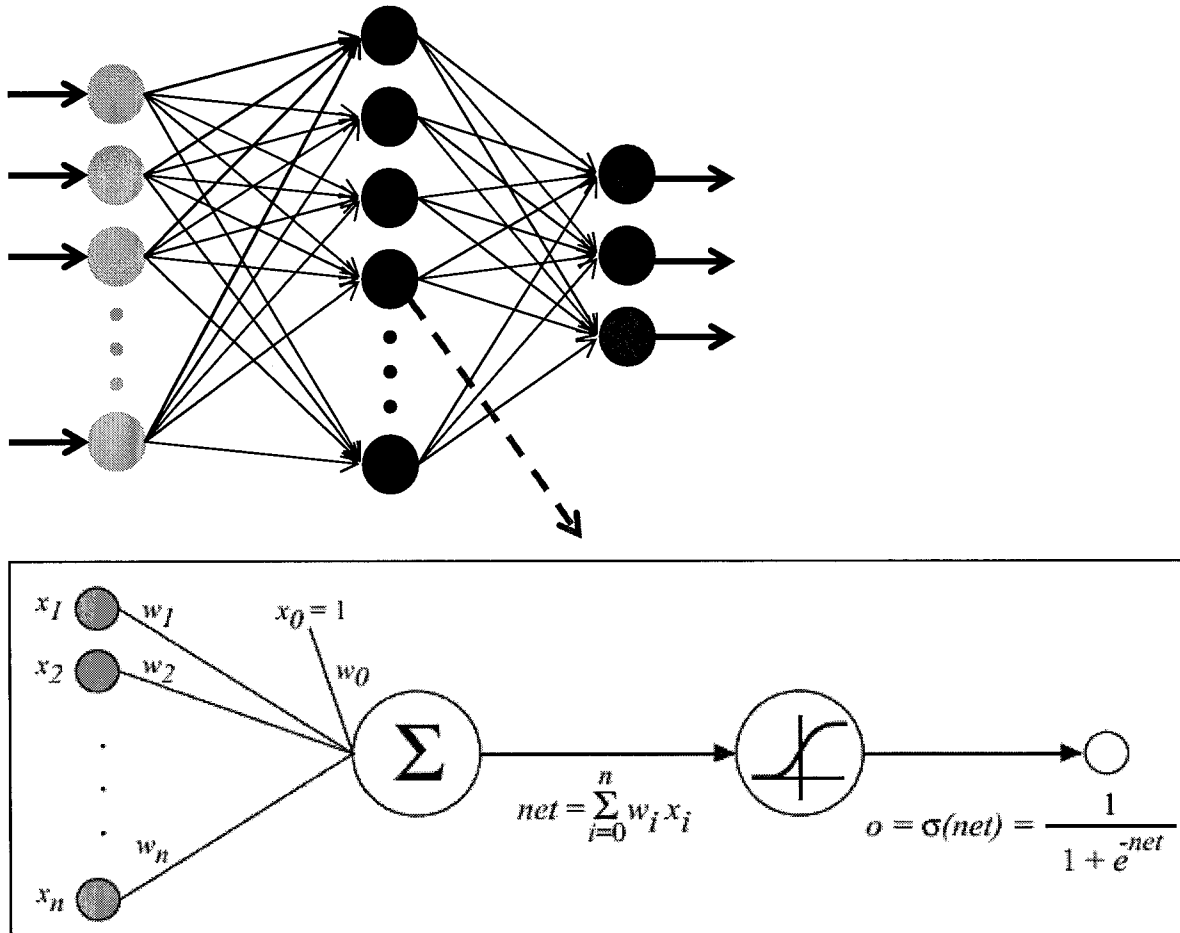


Figure 3. Figure of question 4.2

Question 5. We use decision tree to examine and approve the credit card application. Could you please construct a decision tree according to the given training dataset? Please write down every step of attribute selection. [16]

DATA	SALARY	LIVING AREA	AGE	EDUCATION	LABEL
1	LOW	NT	YOUNG	HIGH	YES
2	LOW	NT	YOUNG	HIGH	YES
3	HIGH	CF	YOUNG	NORMAL	YES
4	LOW	NT	MIDDLE	NORMAL	NO
5	LOW	CF	OLD	HIGH	NO
6	HIGH	CF	MIDDLE	HIGH	YES
7	LOW	CF	YOUNG	HIGH	NO
8	LOW	NT	OLD	NORMAL	NO
9	HIGH	NT	MIDDLE	NORMAL	NO
10	LOW	NT	OLD	NORMAL	NO
11	HIGH	CF	MIDDLE	NORMAL	YES
12	LOW	NT	MIDDLE	NORMAL	NO
13	HIGH	NT	OLD	NORMAL	NO
14	HIGH	CF	OLD	HIGH	YES
15	LOW	NT	MIDDLE	NORMAL	NO
16	HIGH	CF	OLD	HIGH	YES
17	HIGH	CF	MIDDLE	NORMAL	YES
18	HIGH	NT	YOUNG	NORMAL	NO
19	HIGH	CF	OLD	NORMAL	YES
20	HIGH	NT	YOUNG	NORMAL	NO

Question 6. According to the Bayesian Networks shown in Figure 4, please calculate $P(D)$ and $P(A|C)$. [12]

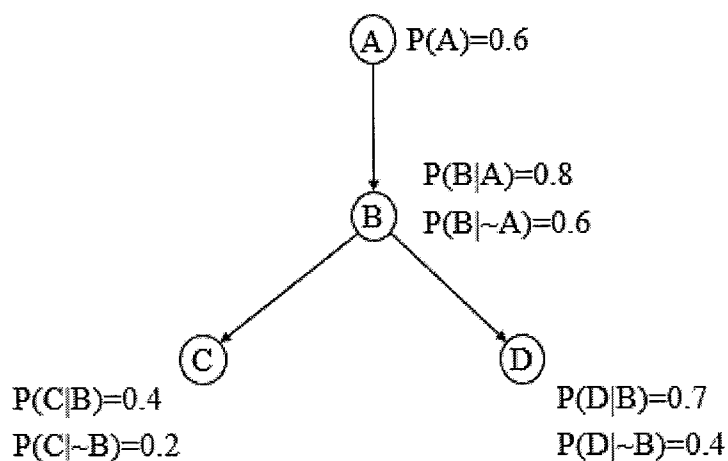


Figure 4. Figure of question 6

** END **